README

Fill-finish site KPI dashboard (Power BI)

A weekly operations dashboard and KPI analysis for a fictional contract manufacturer that fills biologics into vials and prefilled syringes on three aseptic lines.

> **About the data.** The dataset is simulated. It was generated with an AI assistant (Claude) and data problems of the kind found in real plant data were planted on purpose and **not disclosed** to me. Finding them was part of the task. All products, clients, suppliers and people are fictional.

What is in this repo

Path	What it is
`plant_kpi_dashboard.pbix`	The Power BI file
`plant_kpi_dashboard.pdf`	PDF export of every page
`powerbi/01_PowerQuery_all_queries.m`	All Power Query (M) queries: raw loads, fixes, model tables, checks
`powerbi/02_DAX_measures.dax`	Every DAX measure (paste into DAX query view)
`powerbi/03_DAX_check_queries.dax`	Queries that reproduce the reference numbers
`powerbi/reference_numbers/`	Weekly and monthly KPI values to check the build against
`data/`	The 18 CSV extracts and the data dictionary
`screenshots/`	One image per report page

How to open it

1. Clone the repo. Open the `.pbix` in Power BI Desktop (November 2024 or later).
2. Home > Transform data > Edit parameters: set `DataFolder` to the full path of `data\` on your PC (end it with `\`).
3. Close & apply, then Refresh.

Report pages

1. **Weekly ops**: KPIs against target, downtime by cause and equipment, quality by line, product and shift, time per process step. Pick the week top left.
2. **Data checks**: 39 checks with failing-row counts and the decision for each, plus row counts against the dictionary.
3. **KPI dictionary**: formula, DAX measure, unit, sources, inclusions, exclusions, target and owner.
4. **KPI conflicts**: two reasonable definitions per KPI, both values, the one chosen and why.
5. **Issue 1, Issue 2, Issue 3**: one page per finding, with evidence from at least two sources, the impact and the alternatives ruled out.
6. **So what / now what**: action, owner and KPI to watch per finding.

KPI dictionary (short form)

KPI	Formula	Target
Throughput	Good units by fill-end week ÷ (scheduled hours × weekly target ÷ standard week hours)	FL-1 220k, FL-2 180k, FL-3 90k → 150k/week from Mar 2026
Yield	Released good units ÷ theoretical units (rejected batch = 0)	96.5%
First-pass quality	Batches with no re-inspection and no major/critical deviation ÷ batches	92%
Scrap rate	(Line loss + inspection rejects + rejected batches) ÷ theoretical units; € at standard cost	2.0 / 2.5 / 2.0%
Unplanned downtime	Unplanned stop minutes while filling ÷ filling minutes	9 / 10 / 9%
On-time delivery	Orders shipped complete by the committed date ÷ orders due	95–96%

Committed date = first confirmed date, or the new date when the client asked for the change. Full definitions are on the KPI dictionary page.

Data checks: what I found and did

Problem	Rows	Decision
Duplicate batch rows in MES	5	Removed
Process order keys with spaces, lower case or no hyphen	9	Normalised
Missing fill end / QC sample count	4 / 14	Derived from steps / unit balance
Downtime stops ending before they start	5	Start and end swapped
Downtime stops lasting 82–137 h, past the end of their batch	3	Minutes excluded (historian shows the line running)
Double-logged downtime stops	35	Excluded
Stops without a reason code	71	Kept as "Not coded"
Defect code STOP used for two different components	1 code	Split into STOP-V and STOP-S
Shipments without a batch	4	Recovered from order and quantity
Due dates moved after the original date had passed	62	OTD measured on the committed date
Due-date changes without a reason	13	Treated as not client-requested
Release decision dated before the review started	3	Excluded from QA lead time
Deviation batch IDs without zero padding	3	Padded
Historian timestamps in UTC	all	Converted to Copenhagen time
Historian fill weight in mg (FL-1, from May 2026)	524	Divided by 1,000
Historian gap on FL-3 (9–12 Feb 2026)	79 h	Flagged
Production data ends 1 Oct 2026 though the lines were scheduled to 3 Oct	6 line-days	Schedule cut at 1 Oct

The full list (39 checks, including the ones that passed) is on the Data checks page.

Findings

1. **FL-2 stoppering unit after deferred maintenance.** Nine 4-weekly PMs in a row (Nov 2025–Jul 2026) were deferred or cancelled for production priority. Stopper stops tripled, adding about 106 hours of downtime (about 590,000 units of capacity), 10 major deviations, 1 rejected batch (€0.63M) and 10 late orders. Back to normal after the July overhaul.
2. **Supplier C plunger stoppers on FL-3.** Since 16 Mar 2026, batches with Supplier C stoppers show 8× the stopper-defect rate of Supplier B. They account for all 33 re-inspections and 33 deviations and about 19,900 extra rejected syringes (€0.66M).
3. **QA batch-record queue since April 2026.** Waiting time before review tripled with no rise in volume. On-time delivery fell to 28–67% (Apr–Jul) and the queue drove 79 of 132 late orders. Re-dating 34 of them hid this in ERP.

Limitations and asks for data engineering

- Final inspection counts are after re-inspection; store first-pass rejects too.
- Make reason codes, due-date change reasons and the shipment batch mandatory.
- Keep the original promise date as its own ERP field.
- State the time zone of every extract; fix the historian fill-weight unit.
- QA staffing is not in the data, so the cause of the QA queue is inferred, not proven.